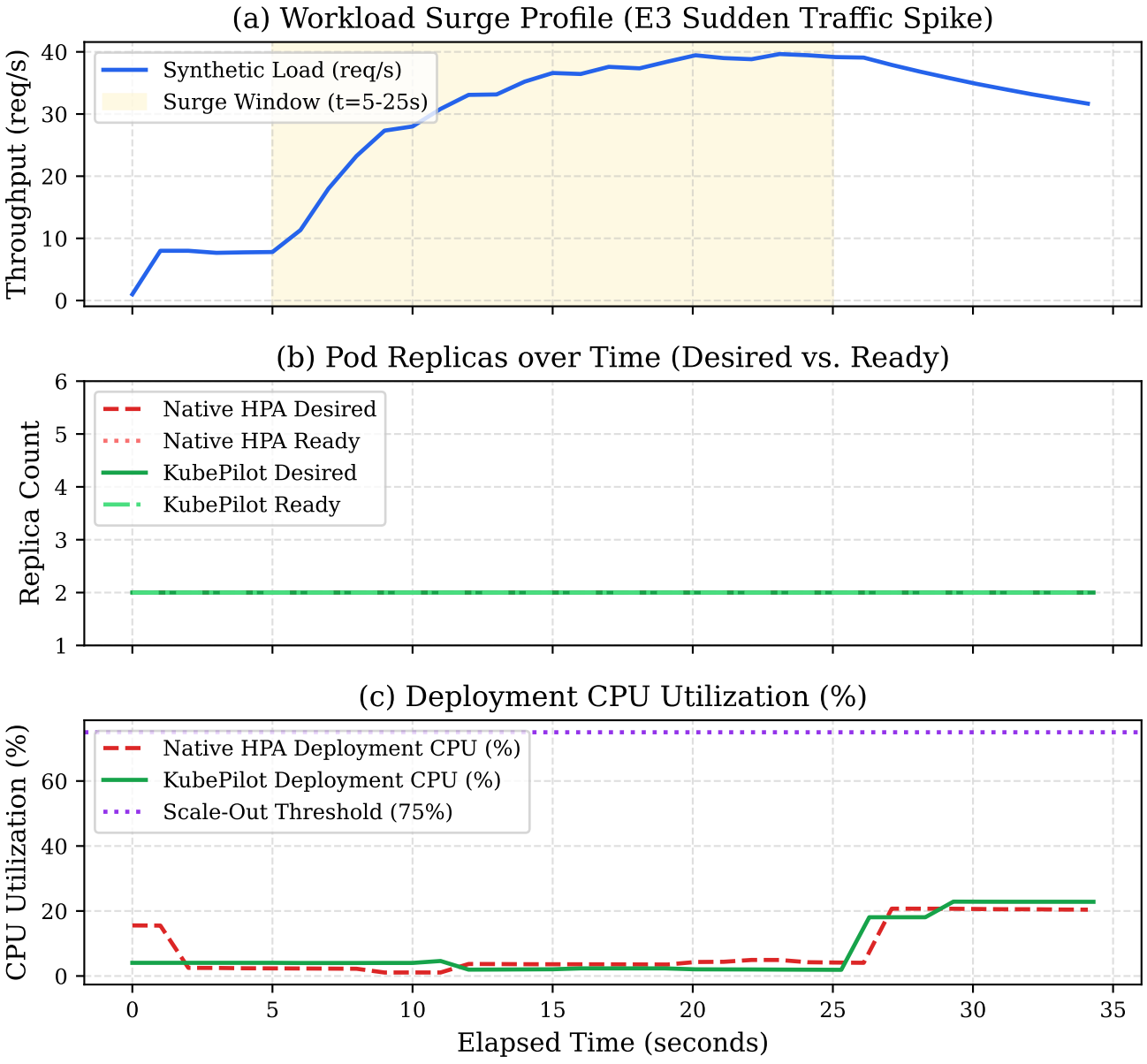
(a) Workload Surge Profile (E3 Sudden Traffic Spike)
Throughput (req/s)
Synthetic Load (req/s)
Surge Window (t=5-25s)
(b) Pod Replicas over Time (Desired vs. Ready)
Replica Count
Native HPA Desired
Native HPA Ready
KubePilot Desired
KubePilot Ready
(c) Deployment CPU Utilization (%)
CPU Utilization (%)
Native HPA Deployment CPU (%)
KubePilot Deployment CPU (%)
Scale-Out Threshold (75%)
Elapsed Time (seconds)